

Griffith School of Engineering  
Griffith University

3004ENG – Project Management Principles

# **Eris TestFlight2 Launch Campaign Project & Team Charter**

Assessment 1: Project and Team Charter.  
Trimester 2, 2026

**Group 1**

10 August 2026

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## Project Team Register

| <b>Name</b>       | <b>Student ID</b> | <b>Discipline</b>        |
|-------------------|-------------------|--------------------------|
| Abdul Wahab Obeid | s5303839          | Mechatronics Engineering |
| Hadi Alkhub       | s5331680          | Software Engineering     |
| Ziyad Alahmari    | s5375826          | Mechanical & Aviation    |
| Ilsa Hashmi       | s5310047          | Software Engineering     |

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# 1 Part A: Team Charter

## 1.1 Team Roles and Responsibilities

To ensure clear delegation of responsibilities and accountability throughout the project lifecycle, project roles have been assigned to each team member. Table 1 outlines each member's project role together with their responsibilities for Assessment 1 (A1) and Assessment 3 (A3), the Group Project Execution Plan (PEP).

Table 1: Team roles and responsibilities.

| Member            | Project Role                       | A1 Responsibilities                                                                            | A3 (PEP) Responsibilities                                                               |
|-------------------|------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Abdul Wahab Obeid | Project Lead & Integration Manager | Overall scheduling & milestone tracking, Draft Purpose & Justification, Draft Success Criteria | Executive Summary, Integration Management, Final QA and document cohesion               |
| Hadi Alkhub       | Lead Planner & Scope Manager       | Draft Team Charter, Communication Plan, Draft SMART Objectives                                 | Scope Management, Schedule Management, Network Logic, Gantt Chart, Schedule Compression |
| Ziyad Alahmari    | Risk & Quality Lead                | Draft Principal Risks, Draft Major Schedule Milestones                                         | Risk Register, Mitigation Strategies, Quality & HSE Governance                          |
| Ilsa Hashmi       | Stakeholder & Resource Lead        | Stakeholder Mapping, High-Level Scope, Maintain MS Teams Workspace                             | Cost Management, Resource Management, Capacity Planning                                 |

**Availability:** Team members are generally available on weekdays either in person at the Nathan campus or remotely via Microsoft Teams. Any temporary availability constraints or scheduling conflicts will be communicated and managed in accordance with the Communication Plan.

## 1.2 Communication Plan

The project uses a hybrid communication strategy, with Microsoft Teams as the single source of truth for correspondence, file management, and version control. To maintain professional boundaries, SMS or external messaging applications will not be used. The following plan outlines procedures to ensure effective collaboration.

### 1.2.1 Meeting Cadence & Attendance

- **Primary Working Session (In-Person):** A two hour meeting preceding each Thursday tutorial at the Nathan campus (Café Rosa) to review progress.
- **Mid-Week Check in (Online):** A mandatory 30 minute online progress sync will occur every Monday at 7:30 pm Brisbane time via Microsoft Teams to resolve schedule breaker tasks and verify handovers.
- **Absence Protocol:** If a team member is unable to attend in person, they must participate in the meeting synchronously via Microsoft Teams.
- **Contingency Debrief:** If remote attendance is impossible, a designated attendee will provide the absent member a debrief within 24 hours, including decisions and action items.

### 1.2.2 Service Level Agreements (SLAs)

- **Routine Queries (General updates):** Maximum 24-hour response window during standard business days (Monday to Friday).
- **Urgent Blockers (Immediate review):** Maximum four-business-hour response time for critical issues flagged using the @Urgent tag within the relevant Microsoft Teams channel.
- **Quality Assurance Buffer:** To facilitate effective peer review, the final draft of any major project deliverable (A1 and A3) must be uploaded to the shared repository at least 48 hours before the official submission deadline. Peer feedback must then be provided within 24 hours of the upload.

### 1.2.3 Enforcement

**SLA Breach Trigger:** Failure to respond within these stated SLA windows, or failure to attend a scheduled cadence meeting without prior written notification, automatically triggers Step 1 of the team's Conflict Resolution Process.

## 1.3 Ground Rules and Team Culture

The following rules establish behavioural expectations and working practices:

### 1.3.1 Team Behaviour and Working Practices

- **Challenge Up:** Team members should constructively challenge ideas and review each other's work to maintain quality.
- **Internal Deadlines:** Internal deadlines are mandatory milestones to allow time for review and quality assurance.

### 1.3.2 Enforceable Contribution

To ensure equitable workload distribution, the following boundaries are mandated:

- **Drafting Compliance:** Internal draft deadlines are mandatory. Missing a deadline by more than 12 hours without prior notice constitutes a breach of the team agreement.
- **Peer Review Mandate:** Every team member must review and provide constructive written feedback on at least two teammates' sections before the final document is compiled.
- **Decision Making:** Major project decisions will be evaluated using a Weighted Scoring Model: Strategic Fit (40%), Technical Feasibility (30%), and Schedule Constraint (30%).

## 1.4 Conflict Resolution Process

Conflicts and SLA breaches will be managed using this progressive process before escalation to the convenor:

### Step 1: Informal Clarification

*Trigger:* A team member exceeds the response SLA by more than 12 hours or misses an internal deadline.

*Action:* The Project Lead (or relevant task lead) contacts the team member to confirm their wellbeing, clarify the reason for the delay, and agree on a revised timeline.

### Step 2: Team Intervention

*Trigger:* No response is received within 24 hours, or a second SLA breach occurs.

*Action:* The team conducts a Microsoft Teams meeting, documents the issue, and agrees on a written recovery plan with clear deliverables and a 48-hour review period.

### Step 3: Tutor Mediation

*Trigger:* The agreed recovery plan is not achieved.

*Action:* The Project Lead provides the documented evidence to the workshop tutor, Muhammad Altaf, and requests mediation. The team member is informed that the issue may affect peer evaluation through Buddycheck.

### Step 4: Convenor Escalation

*Trigger:* The member remains unresponsive following tutor mediation.

*Action:* The matter is escalated to the course convenor for further action and potential assessment impacts.

## 1.5 Fallback Clause and Buddycheck

**Work Redistribution:** If a member withdraws or is excluded:

- Outstanding deliverables are redistributed equally, and the Lead Planner will update the project schedule

**Charter Review:** The team will review this charter in the Week 8 workshop to evaluate performance and revise roles if necessary.

**Buddycheck Peer Assessment:** Individual marks may be adjusted based on Buddycheck peer evaluations. Evidence for fair assessment includes Microsoft Teams records, version history, and task allocations, supporting the Peer Assessment Factor (PAF).

## 1.6 Team Approval and Signatures

By signing this Team Charter, each team member confirms that they have read, understood, and agree to comply with its terms.

### Team Member 1

Name: Abdul Wahab Obeid

Role: Project Lead & Integration Manager

Signature:  \_\_\_\_\_

Date: 08 / 08 / 2026

### Team Member 2

Name: Hadi Alkhub

Role: Lead Planner & Scope Manager

Signature:  \_\_\_\_\_

Date: 08 / 08 / 2026

### Team Member 3

Name: Ziyad Alahmari

Role: Risk & Quality Lead

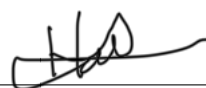
Signature:  \_\_\_\_\_

Date: 08 / 08 / 2026

### Team Member 4

Name: Ilsa Hashmi

Role: Stakeholder & Resource Lead

Signature:  \_\_\_\_\_

Date: 08 / 08 / 2026

## 2 Part B: Project Charter

### 2.1 Background

Following the July 2025 Eris test flight, Gilmour Space Technologies initiated the TestFlight2 Launch Campaign. While TestFlight1 successfully cleared the tower and proved ground infrastructure viability, it was terminated at T+14 seconds due to stability loss (Smith, 2025). Post flight analysis identified thrust loss in two first-stage hybrid motors, caused by electrical and thermal faults in the pump subsystem.

To resolve these technical anomalies and validate the system, Gilmour Space has commissioned the Eris TestFlight2 Launch Campaign. This project is strictly bounded as an operational execution phase. It encompasses cleanroom integration, regulatory safety approvals, pre flight rehearsals, range safety coordination, and the launch itself.

This campaign establishes a reliable Australian sovereign orbital launch capability, serving commercial sponsors, rideshare customers, and defence agencies facing constrained launch options. It respects our operational partnership with the Bowen community and Juru Traditional Custodians (Gilmour Space, 2026). TestFlight2 will evaluate Eris’s capability to safely insert payloads into Low Earth Orbit (LEO) and collect high resolution telemetry to refine future iterations.

### 2.2 Objectives

Aligning with Gilmour Space Technologies’ target to conduct the second Eris test flight by Q1 2027 (Next Spaceflight, 2026), this project’s scope is bounded to Launch Site Operations. The PM is responsible for receiving the pre manufactured vehicle, pad assembly, testing, and launch execution.

- **Objective 1 (Physical Output):** Complete mechanical assembly and electrical integration of the Eris rocket on the Bowen launch pad, verifying all GSE interfaces by October 15, 2026.
- **Objective 2 (Financial Output):** Manage and control all localized Bowen site operations, ground logistics, and equipment rentals strictly within the allocated campaign operational budget, holding an unallocated contingency reserve to absorb weather delays or emergency repairs.
- **Objective 3 (Testing Output):** Successfully conduct an integrated Wet Dress Rehearsal (WDR) and a 5 second Static Fire Test of the four first-stage hybrid engines on the pad, resolving 100% of critical “No-Go” electrical, thermal, or software anomalies by October 30, 2026.
- **Objective 4 (Operational Outcome):** Pass the Launch Readiness Review (LRR) by

demonstrating the pad and vehicle satisfy all safety protocols, securing flight clearance from regulatory authorities by November 10, 2026.

- **Objective 5 (Strategic Benefit):** Execute the final liftoff sequence during the scheduled November 2026 launch window, successfully capturing and decrypting at least 98.5% of high-resolution flight telemetry data to hand over to the engineering teams. This validated flight heritage is required to support the acquisition of future commercial satellite launch contracts and establish reliable Australian sovereign space access.

## 2.3 Scope

### 2.3.1 Inclusions (What is In-Scope)

- **Campaign Management:** Managing localized scheduling, risk registers, and operational budgets through closeout.
- **Pad Integration:** Receiving rocket stages, coordinating vertical stacking, and verifying ground support equipment (GSE) interfaces.
- **Launch Pad Verification:** Operational execution of the Wet Dress Rehearsal (WDR) and the 5-second Static Fire Test to verify propulsion system integrity on the pad.
- **Operational Readiness:** Managing on-site HSE (Health, Safety, and Environment) compliance parameters and coordinating Range Safety to pass the Launch Readiness Review (LRR).
- **Execution & Telemetry Handover:** Initiating launch countdown and handing over the decrypted flight telemetry database to engineering within 5 days post launch.
- **Indigenous Land & Cultural Stewardship:** Operational execution of on-site environmental safeguards and cultural heritage protection monitoring in direct coordination with Juru liaison officers.
- **Local Community Hazard & Safety Management:** Coordinating range clearances and issuing public safety alerts prior to static fires and launch to protect Bowen residents.

### 2.3.2 Exclusions (Deliberately Out-of-Scope)

- **Primary Regulatory Licensing:** Excludes securing the foundational Launch Facility Licence, already granted to Bowen Spaceport in March 2024. (Australian Space Agency, 2025).
- **Upstream Manufacturing & Engineering R&D:** Deliberately excludes the physical redesign, structural manufacturing, or off-site engineering of rocket stages or hybrid engines.

- **Upstream Supply Chain Logistics:** Excludes the physical transport, shipping permits, and freight management of vehicle components to the Bowen spaceport receiving gate.
- **Engineering Defect Resolution:** Excludes resolving root cause design flaws or hardware anomalies identified during pad testing, defective components will be packaged and handed back to the R&D division for resolution.
- **Downstream Commercialization:** Excludes post flight commercial satellite payload operations, long-term orbital tracking, or the marketing, sales, and contracting of future commercial launch campaigns.

## 2.4 Key Stakeholder Analysis and Power Interest Classification

To ensure strategic alignment and robust engagement throughout the campaign, the project’s key stakeholders have been identified, analyzed, and mapped across a standard Power Interest Matrix based on the tactical realities of the Bowen Orbital Spaceport operations:

Table 2: Key Stakeholder Analysis and Classification

| Stakeholder Entity           | Specific Party                                                 | Project Interest / Expectation                                                          | Power | Interest | Engagement Strategy                                                                          |
|------------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------|-------|----------|----------------------------------------------------------------------------------------------|
| Gilmour Executive Sponsor    | Gilmour Space Technology                                       | Validation of Australian sovereign launch capability, and commercial viability of Eris. | High  | High     | <b>Managed Closely:</b> Bi-weekly status updates and milestone gate approvals.               |
| National Space Regulator     | Australian Space Agency                                        | 100% compliance with civil aviation safety and space flight licensing protocols.        | High  | Low      | <b>Keep Satisfied:</b> Proactive regulatory filing and submit safety cases 60 days prior.    |
| Local Indigenous Custodians  | Juru Traditional Owners                                        | Preservation of cultural sites at Abbot Point and environmental land/water protection.  | High  | High     | <b>Managed Closely:</b> Direct consultation and active monitoring of the Juru land.          |
| Regional Town Residents      | Bowen Community & Businesses                                   | Public safety during flight, minimization of closures, and regional economic benefits.  | Low   | High     | <b>Keep Informed:</b> Transparent community boards and publish clearance maps 14 days prior. |
| Primary Customer             | Commercial Payload Sponsor                                     | Reliable orbital insertion of the satellite and payload safety during pad integration.  | Med   | High     | <b>Keep Informed:</b> Weekly integration logs, witness LRR review.                           |
| Local Conservation Authority | Great Barrier Reef Marine Park Authority (Gilmour Space, 2025) | Minimal environmental disruption from rocket exhaust, noise, and marine stage debris.   | High  | Low      | <b>Keep Satisfied:</b> Execute joint operating agreements, share monitoring logs.            |
| Public & Regional Media      | QLD Regional Press                                             | Transparency regarding safety, flight delays, and regional economic impacts.            | Low   | Low      | <b>Monitor:</b> Distribute standard press releases post launch.                              |

## 2.5 Major Schedule Milestones

The following milestones represent the major checkpoints and decision points throughout the project lifecycle.

### Phase 1: Conception Phase

- **Milestone 1 (M-1): Project Charter (A1) Formally Approved**

*Target Date:* August 7, 2026

*Predecessors:* None (Project Commencement Base)

*Deliverable/Acceptance:* Joint signature by all partners.

### Phase 2: Definition Phase

- **Milestone 2 (M-2): Project Execution Plan (PEP - A3) Baseline Locked**

*Target Date:* October 2, 2026

*Predecessors:* M-1

*Deliverable/Acceptance:* Approval of WBS, Gantt schedule, cost baseline, and HSE plan.

**DECISION GATE 1 (Sponsor Sign-off):** Gilmour Space Executive Board reviews the final cost benefit baseline.

### Phase 3: Execution Phase

- **Milestone 3 (M-3): Regulatory Airspace Flight Permits Secured**

*Target Date:* October 25, 2026

*Predecessors:* M-2

*Deliverable/Acceptance:* Formal issue of flight clearances and civil aviation airspace arrangements.

**DECISION GATE 2 (Regulatory Sign-off):** The campaign cannot proceed without active permits in hand.

- **Milestone 4 (M-4): Physical Vehicle Integration & Vertical Pad Stacking Completed**

*Target Date:* October 28, 2026

*Predecessors:* M-2, M-3

*Deliverable/Acceptance:* Delivery of pre manufactured stages to Bowen, vertical stacking of the 23 meter rocket, and verification of GSE electrical interfaces.

- **Milestone 5 (M-5): Integrated Wet Dress Rehearsal (WDR) and Static Fire Completed**

*Target Date:* October 30, 2026

*Predecessors:* M-4

*Deliverable/Acceptance:* Countdown executed to 10 seconds before simulated liftoff,

and a nominal 5-second hybrid motor static fire ignition test completed with zero critical anomalies.

- **Milestone 6 (M-6): Launch Readiness Review (LRR) Clearance Secured**

*Target Date:* November 10, 2026

*Predecessors:* M-5

**DECISION GATE 3 (Range Safety Sign-off):** Final pre flight gate. The Launch Director, Range Safety Officer, and Juru Land Liaisons review site parameters, community sweeps, weather windows, and vehicle diagnostic telemetry.

- **Milestone 7 (M-7): Eris TestFlight2 Liftoff and Orbital Launch**

*Target Date:* November 15, 2026 (Opening of Launch Window)

*Predecessors:* M-6

*Deliverable/Acceptance:* Successful ignition and launch execution; Eris clears Bowen tower, maintains thrust past T+14 second threshold, and achieves target Low Earth Orbit (LEO) coordinates.

#### **Phase 4: Closing Phase**

- **Milestone 8 (M-8): Flight Telemetry Decryption and Engineering Handoff Completed**

*Target Date:* November 22, 2026

*Sequence Predecessors:* M-7

*Deliverable/Acceptance:* Capture and secure transfer of at least 98.5% of high resolution sensor data packets to Gilmour's corporate engineering database.

- **Milestone 9 (M-9): Formal Project Closeout & Lessons Learned Documented**

*Target Date:* December 1, 2026

*Sequence Predecessors:* M-8

*Deliverable/Acceptance:* Submission of Final Campaign Evaluation Report, completion of post flight Bowen range environmental audits, and formal release of campaign personnel back to functional roles.

## 2.6 High-Level Risk Register

To ensure proactive project governance, high-level campaign risks have been decomposed to isolate their triggering conditions and downstream impacts:

Table 3: High-Level Project Risk Register

| Risk ID & Title                                      | Root Cause (Triggering Condition)                                                                         | Direct Project Effect (Specific Impact)                                                                                                                       |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>R-1: Asymmetrical Thrust &amp; Stability Loss</b> | Upstream electrical or thermal insulation faults recurring within the engine’s redesigned pump subsystem. | Asymmetrical thrust loss, vehicle instability, early flight termination, destruction within the safety zone, and catastrophic schedule delay.                 |
| <b>R-2: GBRMPA Permit Breach &amp; Debris</b>        | Launching northeast over the Coral Sea, adjacent to Great Barrier Reef Marine Park boundaries.            | Statutory breach of environmental permits, immediate regulatory suspension of Bowen Spaceport facility license, and severe civil fines.                       |
| <b>R-3: Ground Station Telemetry Loss</b>            | Localized radio frequency interference at the Bowen tracking station antenna array.                       | Complete corruption or loss of real-time flight data stream, failing to deliver the mandatory 98.5% performance database, leaving flight heritage unverified. |
| <b>R-4: Juru Land &amp; Range Friction</b>           | Heavy acoustics, sound pressure levels, or extended beach/marine range exclusions at Abbot Point.         | Active protest, local community friction, or formal complaints to QLD Government, leading to cancellation of launch permit or range clearance.                |
| <b>R-5: GSE Cryogenic Component Freezing</b>         | Pumping extremely cold liquid oxygen (LOX) through the launch pad’s fuel lines and valves during the WDR. | Frozen or cracked lines causing immediate test abort, physical pad equipment damage, and major drain on emergency budget reserves.                            |

## 2.7 Success Criteria

The following success criteria will be used to evaluate whether the project has successfully achieved its intended outcomes upon completion.

### 2.7.1 Project Management Execution Success (Schedule, Cost & Safety)

- **Schedule Variance:** The final Eris TestFlight2 liftoff sequence was executed within the approved launch window before Q1 2027.
- **Cost Variance:** Launch site operations were completed under the baseline budget without requiring additional funding beyond contingency.
- **HSE & Cultural Compliance:** The campaign achieved zero lost-time injuries (LTIs), zero GBRMPA environmental breaches, and zero violations of the Juru Cultural

Heritage Agreement.

- **Scope Boundary Discipline:** The Project Manager prevented scope creep, 100% of hardware defects from the Static Fire Test were returned to R&D rather than triggering unauthorized on pad redesigns.

### 2.7.2 Deliverable & Handover Success (Quality & Technical)

- **Testing Integrity:** The vehicle successfully passed the Wet Dress Rehearsal (WDR) and 5-second Static Fire Test, resolving all critical anomalies to successfully secure Launch Readiness Review (LRR) clearance on the first formal submission.
- **Telemetry Handover:** The team successfully handed over a decrypted database with > 98.5% of flight telemetry to engineering within 5 days of liftoff, providing validated flight heritage to close the project.

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